

GenAI Energy Advisor — Facility Energy Report

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1. Facility Executive Summary

The facility consumed 88079.98 kWh over 30 days, with top opportunities for savings discovered in unscheduled running and tariff optimization. The robotic welder and injection molder machines presented the most significant opportunities for reduction, with potential savings of █122.94 and █1905.74 respectively. A comparison of shift-to-shift waste revealed that Shift A had 15% less waste than Shift C. Implementing these recommendations can lead to substantial financial and carbon savings, with an estimated reduction of 366.85 kg CO2e. By optimizing production schedules and reducing unscheduled running, the facility can minimize energy waste and lower costs. The recommended changes can be implemented with varying levels of difficulty, requiring collaboration between operators, production planners, and facility managers. Overall, addressing these opportunities will contribute to a more efficient and sustainable facility operation.

Metric	Value	Notes
Total Facility Consumption	88,080.0 kWh	Sum of all metered machines
Total Idle Energy	56,877.6 kWh	64.6% of total draw
Financial Cost of Idle State	█451,595.18	Calculated using dynamic tariff bands
Carbon Emissions saved (Potential)	46,639.7 kg CO2e	Offset at regional grid factor (0.82 kg/kWh)

2. Machine-Wise Energy Dashboard

Machine ID	Active (kWh)	Idle (kWh)	Idle %	Cost of Idle	Status
CNC_Mill_1	4,585.1	8,794.5	65.7%	█73,240	Critical
CNC_Lathe_1	2,242.5	4,718.6	67.8%	█40,694	Critical
Robotic_Welder_1	2,000.5	6,080.2	75.2%	█45,075	Critical
Pick_Place_1	828.8	1,454.2	63.7%	█12,393	Critical
Assembly_Conveyor_1	1,040.7	1,826.7	63.7%	█15,713	Critical
Injection_Molder_2	12,294.2	19,887.1	61.8%	█143,553	Critical
Thermoformer_2	5,388.3	8,360.2	60.8%	█71,497	Critical
Packaging_Robot_2	1,325.5	2,602.5	66.3%	█22,336	Critical
Labeling_Machine_2	454.4	864.4	65.5%	█7,472	Critical
Palletizer_2	1,042.4	2,289.3	68.7%	█19,622	Critical

3. GenAI Recommendation Cards

#1 Robotic_Welder_1 — Unscheduled Running

Detected Issue: Excess power draw of +1.66 kW during unscheduled production

Action: Schedule production to minimize unscheduled running

Weekly Savings: 18.02 kWh | ■117.13 | 14.78 kg CO₂e

Owner: Production Planner | **Difficulty:** Medium | **Confidence:** 80%

Evidence: Machine utilization data, Production schedule analysis

Operator Instructions (Step-by-Step): 1. Review production schedule, 2. Identify unscheduled running, 3. Adjust schedule to minimize unscheduled running

Management Perspective: Implementing this recommendation will result in estimated weekly savings of ■117.13 and a reduction of 14.78 kg CO₂e

#2 CNC_Mill_1 — Unscheduled Running

Detected Issue: Excess power draw during unscheduled production running

Action: Schedule production to minimize unscheduled running

Weekly Savings: 12.73 kWh | ■127.26 | 10.44 kg CO₂e

Owner: Operator | **Difficulty:** Low | **Confidence:** 70%

Evidence: Machine utilization data, Production schedule analysis

Operator Instructions (Step-by-Step): 1. Review production schedule, 2. Identify unscheduled running, 3. Adjust schedule to minimize unscheduled running

Management Perspective: Implementing this recommendation will result in estimated weekly savings of ■127.26 and a reduction of 10.44 kg CO₂e

#3 Injection_Molder_2 — Renewable Shift

Detected Issue: Operational load scheduled during high-tariff or low-solar window

Action: Schedule production during low-tariff or high-solar window

Weekly Savings: 476.43 kWh | ■1667.52 | 366.85 kg CO₂e

Owner: Production Planner | **Difficulty:** Medium | **Confidence:** 90%

Evidence: Tariff data, Solar irradiance data

Operator Instructions (Step-by-Step): 1. Review production schedule, 2. Identify opportunities for renewable shift, 3. Adjust schedule to optimize energy usage

Management Perspective: Implementing this recommendation will result in estimated weekly savings of ■1667.52 and a reduction of 366.85 kg CO₂e

#4 Injection_Molder_2 — Tariff Optimization

Detected Issue: Operational load scheduled during high-tariff or low-solar window

Action: Schedule production during low-tariff or high-solar window

Weekly Savings: 317.62 kWh | ■1905.74 | 0.0 kg CO₂e

Owner: Operator | **Difficulty:** Low | **Confidence:** 80%

Evidence: Tariff data, Solar irradiance data

Operator Instructions (Step-by-Step): 1. Review production schedule, 2. Identify opportunities for tariff optimization, 3. Adjust schedule to optimize energy usage

Management Perspective: Implementing this recommendation will result in estimated weekly savings of ■1905.74

4. 4-Week Implementation Roadmap

Week	Actions	Responsibility
Week 1	Address CNC_Mill_1 weekend shutdown procedure & check welder contactor loop.	Operator / Maintenance
Week 2	Configure automated standby parameters (30 min idle) on packaging robots & conveyors.	Maintenance
Week 3	Train planners on shifting CNC Mill heavy test cycles to Solar Peak window.	Production Planner
Week 4	Establish daily auditing of dynamic rolling thresholds on dashboard.	Facility Manager

5. Appendix: Methodology & Glossary

Methodology: Data is ingested from local IoT meters at 15-minute intervals. Schedules are cross-referenced from the MES schedule files. Dynamic baselines are computed continuously using a 14-day rolling mean and standard deviation. Anomaly categorization incorporates multi-modal sensor checks.

Glossary: *kW Demand*: Peak power draw in kilowatts; *Power Factor (PF)*: Ratio of real power to apparent power (higher is better); *Grid Intensity*: Average CO2 emissions per kWh generated.